Practice Exam - 1

Question 3 of 117

The patient is a 66-year-old woman with trastuzumab-induced cardiomyopathy (LVEF: 40%, NYHA class III, Stage C). Her past medical history is significant for atrial fibrillation, type 2 diabetes, hypertension, and depression.

Her medications include lisinopril 20 mg daily, furosemide 40 mg twice daily, carvedilol 12.5 mg twice daily, spironolactone 25 mg daily, and digoxin 0.125 mg daily, apixaban 5 mg twice daily, and metformin 1000 mg twice daily. She has been stable on these medications for the past month. Her most recent laboratory results include sodium (Na) 140 mEq/L, potassium (K) 4.0 mEq/L, chloride (Cl) 105 mEq/L, bicarbonate 26 mEq/L, blood urea nitrogen (BUN) 12 mg/dL, serum creatinine (SCr) 0.8 mg/dL, creatinine clearance (CrCL) 55 ml/min, glucose 70 mg/dL, calcium 9.0 mg/dL, phosphorus 2.8 mg/dL, magnesium 2.0 mEq/L, and digoxin 0.9 ng/mL. Her vital signs today include blood pressure 112/70 mm Hg and heart rate 68 beats/minute. Her lung examination is clear.

She will be admitted for elective hip surgery.

Which of the following should be done prior to surgery?

A. Stop the apixaban two days prior to surgery

B. Stop the metformin seven days prior to surgery

C. Stop carvedilol three days prior to surgery

D. Stop atorvastatin two days prior to surgery

Commentary | References

Testing Point: The Board-certified AHFTC specialist should have an understanding of perioperative management of a patient with HFrEF in order to reduce the risk of adverse events.

Answer: A. Stop the apixaban two days prior to surgery. Major adverse cardiac events (MACE) are some of the most common complications occurring in the perioperative period, particularly for patients with heart failure. The use of the direct acting oral anticoagulants (DOACs) are now recommended over warfarin by the AHA/ACC Atrial Fibrillation management guidelines for therapeutic anticoagulation.

As more than 35% of patients with heart failure have concomitant atrial fibrillation, management of anticoagulation in the perioperative setting is important to minimize MACE in this population. When considering discontinuation of a DOAC, one must balance thromboembolic risk, renal function, and bleeding risk of the surgery. This patient has a CHADs- Vasc score of 4, which is moderate-high risk of stroke. Her renal function is adequate as her CrCl > 30 ml/ min, but the surgery is considered a high bleeding-risk procedure. With this in mind, apixaban should be stopped 48 hours prior to surgery based on the PAUSE study, making Answer A the best choice. If this surgery was low risk for bleeding, stopping the apixaban one day prior would be acceptable. Due to the risk of lactic acidosis, stopping metformin will be important; however, stopping seven days prior to surgery is excessive and could lead to hyperglycemia prior to surgery. The American Diabetes Association recommends withholding metformin the day of surgery, particularly in patients with a CrCL > 30 ml/min, making Answer B a less optimal choice. Sudden discontinuation of carvedilol can result in rebound hypertension, tachycardia and potential worsening of heart failure.

The 2014 ACC/AHA perioperative practice guidelines for non-cardiac surgery recommend beta blockers and statins be continued in patients who are on these medications chronically (1B recommendations for both).

Thus, Answers C and D are not the best choice.